JACC VOL. ■, NO. ■, 2026

Implications of Modest Reductions in ABI Measurements for Risk of Major Adverse Limb Events Among 223,350 Veterans

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ABSTRACT

BACKGROUND Low resting ankle-brachial index (ABI) is a marker of poor cardiovascular outcomes. Whether modest decrements in ABI translate into higher risks of major adverse limb events (MALE) is unclear because prior studies are small or lack racial diversity and incident outcomes.

OBJECTIVES The purpose of this study was to assess the association between the full range of ABI measures and incident MALE in a large, diverse cohort.

METHODS Using data from the Veterans Aging Cohort Study-National Cohort, we analyzed a prospective, longitudinal cohort of veterans free of prevalent peripheral artery disease (PAD). Participants were enrolled beginning January 1, 2000, and followed through September 30, 2021. The exposure was resting ABI (continuous and categorical), and the primary outcome was MALE, defined as amputation or revascularization using administrative codes. Secondary outcomes included total amputation, major amputation, or revascularization. Cox proportional hazards models assessed the overall association between ABI and MALE and stratified by sex or race. Models were adjusted for demographics and PAD risk factors.

RESULTS The analysis included 223,350 people, including 8,207 women and 42,173 Black individuals. There were 28,191 MALE. Risk of MALE followed an inverse j-shaped distribution across the continuous ABI spectrum. Compared with a categorical ABI of 1.11 to 1.20, borderline ABI values (range 0.91-1.00) were associated with an increased risk of MALE in the total population as well as sex/race subgroups: total population: HR: 1.53 (95% CI: 1.43-1.64); men: HR: 1.53 (95% CI: 1.43-1.64); women: HR: 2.00 (95% CI: 1.18-3.38); White individuals: HR: 1.60 (95% CI: 1.48-1.73); Black individuals: HR: 1.39 (95% CI: 1.18-1.64). Similar associations were demonstrated for major amputation (HR: 1.34 [95% CI: 1.17-1.54]), total amputation (HR: 1.22 [95% CI: 1.12-1.33]), and revascularization (HR: 2.05 [95% CI: 1.87-2.26]) in the full cohort. ABI was a stronger marker of MALE risk than established risk factors, including current smoking and prevalent cardiovascular disease.

CONCLUSIONS In a large, diverse cohort free of prior PAD, ABI values across the full spectrum were associated with an increased risk of incident MALE, suggesting that treating the ABI as a binary measure does not adequately capture clinical risk. Further studies are needed to better understand why MALE occurs despite near-normal ABI values. (JACC. 2026;■:■-■) © 2026 the American College of Cardiology Foundation. Published by Elsevier. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received August 30, 2025; revised manuscript received March 25, 2026, accepted April 2, 2026.

ISSN 0735-1097/$36.00 https://doi.org/10.1016/j.jacc.2026.04.004

ABBREVIATIONS AND ACRONYMS

CVD = cardiovascular disease

eGFR = estimated glomerular filtration rate

MALE = major adverse limb events

PAD = peripheral artery disease

VA = Veterans Affairs

VHA = Veterans Healthcare Administration

The ankle-brachial index (ABI) is the primary diagnostic test for peripheral artery disease (PAD). This hemodynamic measure is defined as the higher brachial artery systolic pressure divided by the higher of either the dorsalis pedis or posterior tibial artery pressure in each leg. Under normal conditions, the arterial pressure at the ankle should be slightly higher than in the arms.[1] An ABI value ≤0.90 is diagnostic of PAD, and a value of 0.91 to 1.00 is classified as borderline.[2] Beyond its value in detecting PAD, an abnormal ABI has been linked to an increased risk of incident coronary events,[3,4] heart failure,[5,6] and cardiovascular mortality.[3,7,8]

Despite these findings, the association between a low ABI and adverse limb outcomes is less clear. Some prior data have shown an association between low ABI and chronic limb-threatening ischemia (CLTI),[9] lower extremity arterial revascularization,[10] or amputation.[9,10] These analyses have been limited by small sample size[10] or number of events,[9,10] lack of racial diversity,[10] single limb ABI measurement,[9] restriction to those with known PAD,[10,11] or restriction to extreme ABI values.[10]

The goal of this study is to examine the association between the full spectrum of ABI measures, including borderline values, and limb outcomes among a large, racially diverse cohort of U.S. veterans. We further examined these associations separately among men and women and Black and White individuals and evaluated subsets of PAD outcomes, including lower extremity major amputation, total amputation, and revascularization.

METHODS

STUDY POPULATION. The study population consisted of participants in the VACS (Veterans Aging Cohort Study)-National Cohort, which consists of all ∼13.5 million veterans receiving care at a Veterans Health Administration (VHA) facility since 2000.[12-14] We utilized electronic health record data obtained from these individuals from January 1, 2000, through September 30, 2021. Eligible participants had at least 1 inpatient or outpatient medical encounter during this timeframe. This study was approved by the Institutional Review Boards of Yale University, the Veterans Affairs (VA) Connecticut Healthcare System, the VA Tennessee Valley Health Care System, and Vanderbilt University Medical Center. It has been granted a waiver of informed consent and is compliant with the Health Insurance Portability and Accountability Act.

The analysis consisted of all VACS-National participants who underwent ABI testing within the VA Health Care System from January 1, 2000, through March 30, 2021 (n = 361,854). After excluding participants with a resting ABI in only 1 limb (n = 78,395) or a missing ABI date or laterality (n = 11,759), the remaining population with a qualifying ABI was 271,700. We further excluded those with a clinical history of PAD defined by the presence of VHA, VHA fee-for-service, Medicaid, or Medicare International Classification of Disease-Ninth/Tenth Revision, or current procedural terminology codes (Supplemental Table 1) (n = 21,264),[15] those who experienced a primary endpoint within 6 months of the qualifying ABI (n = 19,058), and those who died or were censored within 6 months of the qualifying ABI (n = 8,028). The final study population was 223,350 (Supplemental Figure 1). Baseline was defined as date of first qualifying ABI with follow-up starting 6 months later to mitigate potential protopathic bias by excluding prevalent PAD. Therefore, our study population consists of all patients who survived and had no MALE events 6 months after the qualifying ABI measurement. In a sensitivity analysis, we began follow-up on the date of qualifying ABI, still excluding those with prevalent PAD at that time point.

EXPOSURE. The primary exposure was the ABI performed within the VHA. For the qualifying ABI study, individuals were classified by whichever limb measure was furthest from the normal value of 1.1. ABI data were extracted directly from the electronic medical record using a validated VHA-developed natural language processing tool.[16]

OUTCOMES. The primary outcome was MALE, a composite endpoint comprising all nontraumatic lower extremity amputations as well as either surgical or percutaneous lower extremity artery revascularization. These events were defined by the presence of ≥1 inpatient and/or ≥2 outpatient VHA, VHA fee-for-service, or Medicare International Classification of Disease-Ninth/Tenth Revision codes, or ≥1 current procedural terminology codes (Supplemental Table 1).[17] We excluded trauma-associated amputations, defined by the presence of administrative codes (Supplemental Table 2) present within 30 days of an otherwise qualifying event. Secondary endpoints included lower extremity nontraumatic above-the-ankle major amputation, total amputation, or surgical/percutaneous lower extremity revascularization separately.

DATA COMPLETENESS AND FOLLOW-UP. For this study, we utilize VA data as well as Program Integrity Tool and fee-for-service data from the VA Corporate Data Warehouse and the VA Informatics and Computing Infrastructure sources. Medicare data is obtained from the VA Information Resource Center. Both Medicare and as Program Integrity Tool/fee-for-service data are incorporated into this study to account for MALE that occurs outside of the VA health care system. These data are linked by scrambled social security codes. Incorporating these 3 sources of data allows for event data to include outcomes from both within and outside the VA health care system. Analyses were truncated at September 30, 2021, to coincide with the end of available linked Medicare data files.

COVARIATES. Our analysis included the following covariates: age, sex, race/ethnicity, Veterans Integrated Service Network,[18] smoking (never/former/current), body mass index, systolic and diastolic blood pressure, antihypertensive medication prescription, estimated glomerular filtration rate, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, triglycerides, statin prescription, antiplatelet therapy prescription (clopidogrel, prasugrel, ticagrelor, or dipyridamole), prevalent cardiovascular disease (CVD) (1 inpatient or 2 outpatient administrative codes for acute myocardial infarction, ischemic stroke, or coronary revascularization), and prevalent diabetes mellitus. We chose these covariates based on a priori data linking them to incident PAD.[19]

All covariates were determined from the VA Corporate Data Warehouse where data from VHA electronic health records are stored. There is high concordance between self-identified race and ethnicity with that reported in the Corporate Data Warehouse.[20,21] Race and ethnicity were categorized as Black, White, or other, which included those identified as American Indian or Alaskan Native, Asian, Native Hawaiian, or Pacific Islander and those of Hispanic ethnicity. Exploratory analyses focused on individuals of Black or White race because of the small number of individuals in the “other” category. Medication use was ascertained using VHA outpatient prescription data, with over-the-counter use of aspirin unavailable in the cohort. Prevalent diabetes was defined using a validated metric comprised of administrative codes, medications, and laboratory values,[22] and smoking status was defined based on validated health risk factor data within the VHA electronic medical record.[23] All covariates were defined using data closest to the qualifying ABI, either before or up to 180 days after this date as previously described.[22,24]

STATISTICAL ANALYSIS. Descriptive analyses compared baseline characteristics by sex or race/ethnicity using the Wilcoxon rank sum test for continuous variables and the chi-square test for categorical variables. Age-adjusted incidence rates for all outcomes were calculated using Poisson regression for the total population as well as after stratification by sex or race. In these analyses, ABI was modeled as a categorical variable with levels ≤0.5, 0.51-0.6, 0.61-0.7, 0.71-0.8, 0.81-0.9, 0.91-1, 1.01-1.1, 1.11-1.2, 1.21-1.3, and >1.3. The incidence rates of MALE, major amputation, total amputation, and revascularization were reported for each ABI category overall and stratified by sex or race with age fixed at 60 years, which was close to the median age of the total study population. The primary outcome and secondary outcomes were censored at the end of study (September 30, 2021), maximum follow-up time of 10 years, last medical contact, or death, whichever happened first.

To help make our results clinically tangible, the 10-year cumulative incidence rates per 1,000 persons with 95% CIs were also reported. The cumulative incidence of having a MALE at 10 years of follow-up was computed using the Aalen-Johansen estimate with death as a competing risk.[25] Propensity score weighting was used to make each of the ABI categories comparable with the cohort as a whole in terms of age, sex, and race distributions.[26] CIs were calculated using quantile bootstraps with 1,000 iterations. The cumulative incidence of 10-year all-cause mortality before MALE was computed using the same methodology. Excess event rates were calculated as the difference in 10-year cumulative incidence from the referent level ABI 1.11 to 1.20 with CIs calculated by computing the difference within each bootstrap.

Cox proportional hazards regression was used to estimate the association of the primary or secondary outcomes with the primary exposure, ABI, modeled as a continuous or as a categorical variable to facilitate the comparison of our results to previous studies. Our primary goal was to estimate a cause-specific hazard of ABI with respect to MALE rather than the association of ABI with cumulative risk of MALE. Therefore, although MALE was censored at death, we used Cox proportional hazards regression instead of Gray and Fine competing risk regression.[27,28] The HRs and their 95% CIs were reported as functions of the ABI values. All models were adjusted for the covariates described in the previous text. Continuous variables were modeled using restricted

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TABLE 1 Baseline Characteristics of Study Participants

	Total (N = 223,350)[a]	Men (n = 215,143)[b]	Women (n = 8,207)[c]	White (n = 165,975)[d]	Black (n = 42,173)[e]	Other (n = 5,162)[f]
Age, y	63 (58-68)	63 (58-68)	58 (53-64)	63 (58-69)	61 (56-67)	63 (58-68)
Male	232,557 (96.3)	–	–	160,771 (96.9)	39,766 (94.3)	4,910 (95.1)
Race						
White	165,975 (74.3)	160,771 (74.7)	5,204 (63.4)	–	–	–
Black	42,173 (18.9)	39,766 (18.5)	2,407 (29.3)	–	–	–
Other[g]	5,162 (2.3)	4,910 (2.3)	252 (3.1)	–	–	–
Smoking status						
Current	99,067 (44.4)	95,837 (44.5)	3,230 (39.4)	73,467 (44.3)	19,437 (46.1)	2,070 (40.1)
Former	68,084 (30.5)	66,320 (30.8)	1,764 (21.5)	52,737 (31.8)	11,057 (26.2)	1,553 (30.1)
Never	45,929 (20.6)	43,018 (20.0)	2,911 (35.5)	32,456 (19.6)	10,227 (24.3)	1,290 (25.0)
Diabetes	106,693 (47.9)	103,818 (51.7)	2,875 (35.0)	77,682 (46.8)	22,004 (52.2)	2,743 (53.1)
Prevalent CVD	62,801 (28.8)	61,412 (28.5)	1,389 (16.9)	48,675 (29.3)	10,288 (24.4)	1,351 (26.2)
Ankle-brachial index	0.96 (0.70-1.16)	0.96 (0.70-1.16)	1.00 (0.82-1.16)	0.97 (0.71-1.17)	0.91 (0.67-1.08)	1.00 (0.75-1.18)
SBP, mm Hg	131 (122-140)	131 (123-140)	128 (120-137)	131 (122-139)	132 (124-141)	130 (122-139)
DBP, mm Hg	75 (70-81)	75 (70-81)	75 (70-81)	75 (69-80)	77 (72-83)	75 (70-81)
Antihypertensive therapy	206,768 (92.8)	199,762 (92.9)	7,006 (85.4)	153,947 (92.8)	39,467 (93.6)	4,743 (91.9)
BMI, kg/m^2	29.7 (25.8-34.2)	29.6 (25.8-34.2)	30.1 (25.7-35.2)	29.8 (26.1-34.4)	28.9 (25.1-33.4)	29.7 (25.9-34.3)
Total cholesterol, mg/dL	164 (138-194)	163 (137-193)	188 (161-218)	163 (137-193)	166 (140-196)	164 (137-196)
LDL cholesterol, mg/dL	91 (70-117)	90 (69-116)	106 (84-133)	90 (69-116)	94 (72-120)	91 (69-118)
HDL cholesterol, mg/dL	40 (34-49)	40 (33-49)	51 (41-62)	39 (33-48)	44 (36-54)	40 (34-49)
Triglycerides, mg/dL	133 (91-200)	134 (91-200)	123 (85-184)	140 (96-209)	108 (76-160)	135 (93-204)
Statin therapy	170,737 (76.7)	165,493 (76.9)	5,244 (63.9)	129,120 (77.8)	30,890 (73.2)	3,929 (76.1)
Antiplatelet therapy	50,546 (23.9)	49,450 (23.0)	1,096 (13.4)	39,874 (24.0)	7,800 (18.5)	1,064 (20.6)
eGFR, mL/min/1.73 m^2	80 (65-95)	80 (65-95)	79 (66-94)	79 (64-93)	85 (68-101)	78 (62-92)

Values are median (Q1-Q3) or n (%). Between group differences for sex or race/ethnicity were assessed using the Wilcoxon rank-sum test or Kruskal-Wallis test for continuous data and the chi-square test for categorical data. Sex and race/ethnicity groups were different in a statistically significant manner ($P < 0.05$) for all covariates except for diastolic blood pressure ($P = 0.16$) and estimated glomerular filtration rate (eGFR) ($P = 0.62$) between sexes. [a]Number missing: 10,040 (race/ethnicity); 10,270 (smoking status); 2,386 (systolic blood pressure [SBP]); 2,386 (diastolic blood pressure [DBP]); 1,930 (body mass index [BMI]); 5,958 (total cholesterol); 7,287 (low-density lipoprotein [LDL] cholesterol); 6,742 (high-density lipoprotein [HDL] cholesterol); 6,934 (triglycerides); 2,721 (eGFR). [b]Number missing: 9,696 (race/ethnicity); 9,968 (smoking status); 2,307 (SBP); 2,307 (DBP); 1,847 (BMI); 5,736 (total cholesterol); 7,018 (LDL cholesterol); 6,491 (HDL cholesterol); 6,674 (triglycerides); 2,575 (eGFR). [c]Number missing: 344 (race/ethnicity); 302 (smoking status); 79 (SBP); 79 (DBP); 83 (BMI); 222 (total cholesterol); 269 (LDL cholesterol); 251 (HDL cholesterol); 260 (triglycerides); 146 (eGFR). [d]Number missing: 7,315 (smoking status); 1,404 (SBP); 1,404 (DBP); 1,133 (BMI); 3,968 (total cholesterol); 4,835 (LDL cholesterol); 4,435 (HDL cholesterol); 4,584 (triglycerides); 1,790 (eGFR). [e]Number missing: 1,452 (smoking status); 371 (SBP); 371 (DBP); 236 (BMI); 966 (total cholesterol); 1,254 (LDL cholesterol); 1,153 (HDL cholesterol); 1,184 (triglycerides); 386 (eGFR). [f]Number missing: 249 (smoking status); 68 (SBP); 68 (DBP); 57 (BMI); 150 (total cholesterol); 172 (LDL cholesterol); 164 (HDL cholesterol); 168 (triglycerides); 71 (eGFR). [g]Other race and ethnicity included individuals identified as American Indian or Alaskan Native, Asian, Native Hawaiian, or Pacific Islander, as well as those of Hispanic ethnicity.

CVD = cardiovascular disease.

cubic splines with up to 4 knots.[29] Similar analyses examined risk of lower limb amputation or revascularization. Missing variables were imputed using predictive mean matching and multiple imputation with chained equations,[30,31] resulting in 10 imputed data sets. The results from analyzing these 10 data sets were combined using Rubin's rule.[32] Association with each endpoint was reported as a multivariable-adjusted HR with 95% CIs. When modeled as a continuous variable, an ABI of 1.1 was used as reference, and when modeled as a categorical variable, an ABI of 1.11 to 1.2 was used as reference.[3,9] In supplemental analyses, variable importance for association with events of interest was ranked using chi-square minus df. Schoenfeld residuals were used to confirm that the proportional hazards assumption was not violated. The type I error rate was set at 0.05. All analyses were performed using R version 3.6.2.[33]

RESULTS

Among 223,350 individuals in the analysis, 8,207 (3.7%) were women and 232,557 (96.3) were men (Table 1). The median age was 63 years (Q1-Q3: 58-69 years) Most were White (165,975; 74.3%) followed by Black (42,173; 18.9%) or other race/ethnicity (5,162; 2.3%). Known PAD risk factors were common, including current smoking (99,067; 44.4%), diabetes (106,693; 47.9%), and prevalent CVD (62,801; 28.8%). The median baseline ABI was 0.96 (Q1-Q3: 0.70-1.16). There were 28,191 incident MALEs over a median follow-up period of 4.3 years, including 5,571 major amputations, 11,259 total amputations, and 21,961 lower extremity revascularizations.

Age-adjusted incidence rates of MALE increased as ABI category decreased from the reference range of 1.11 to 1.20 in the study population (Supplemental

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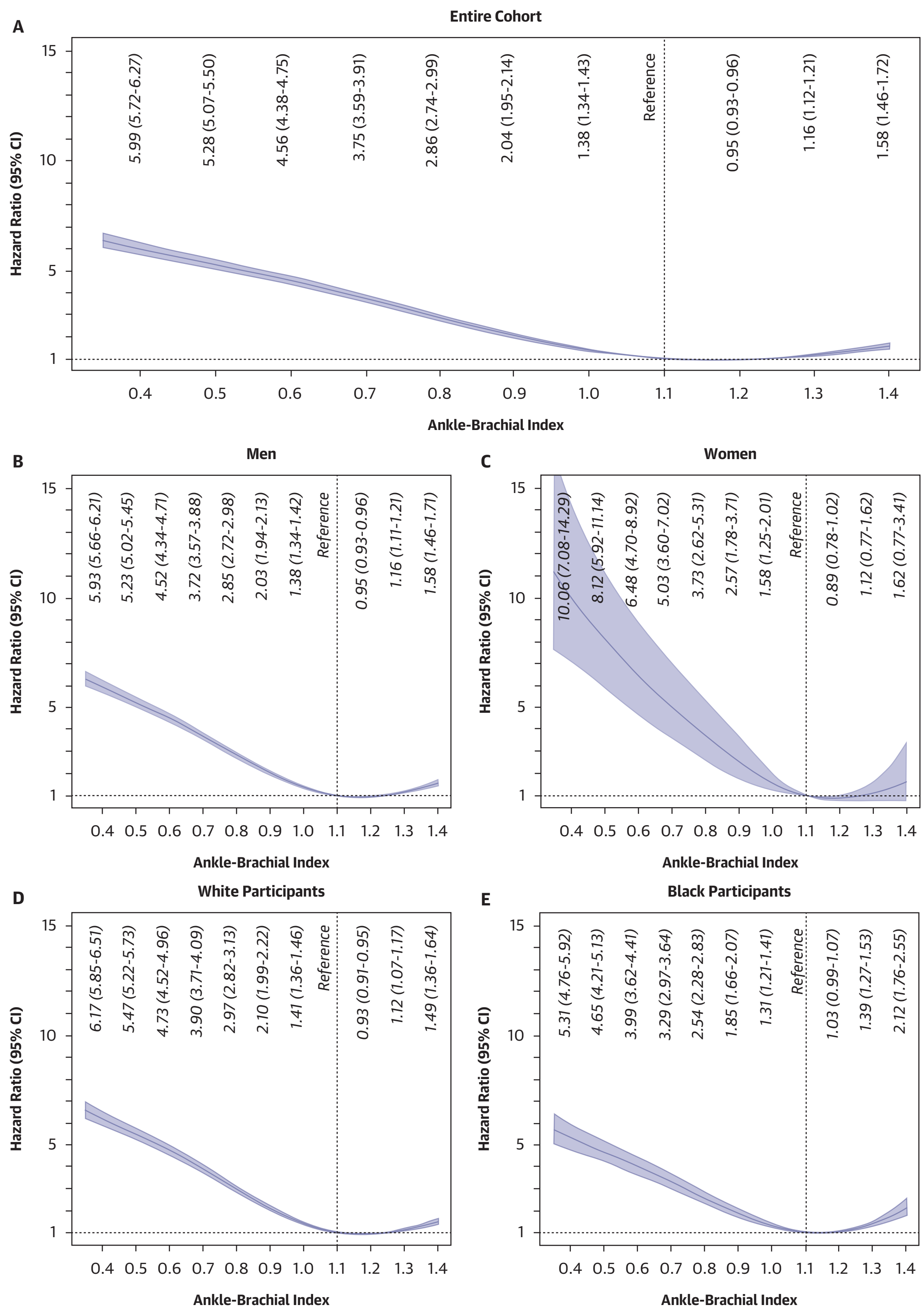
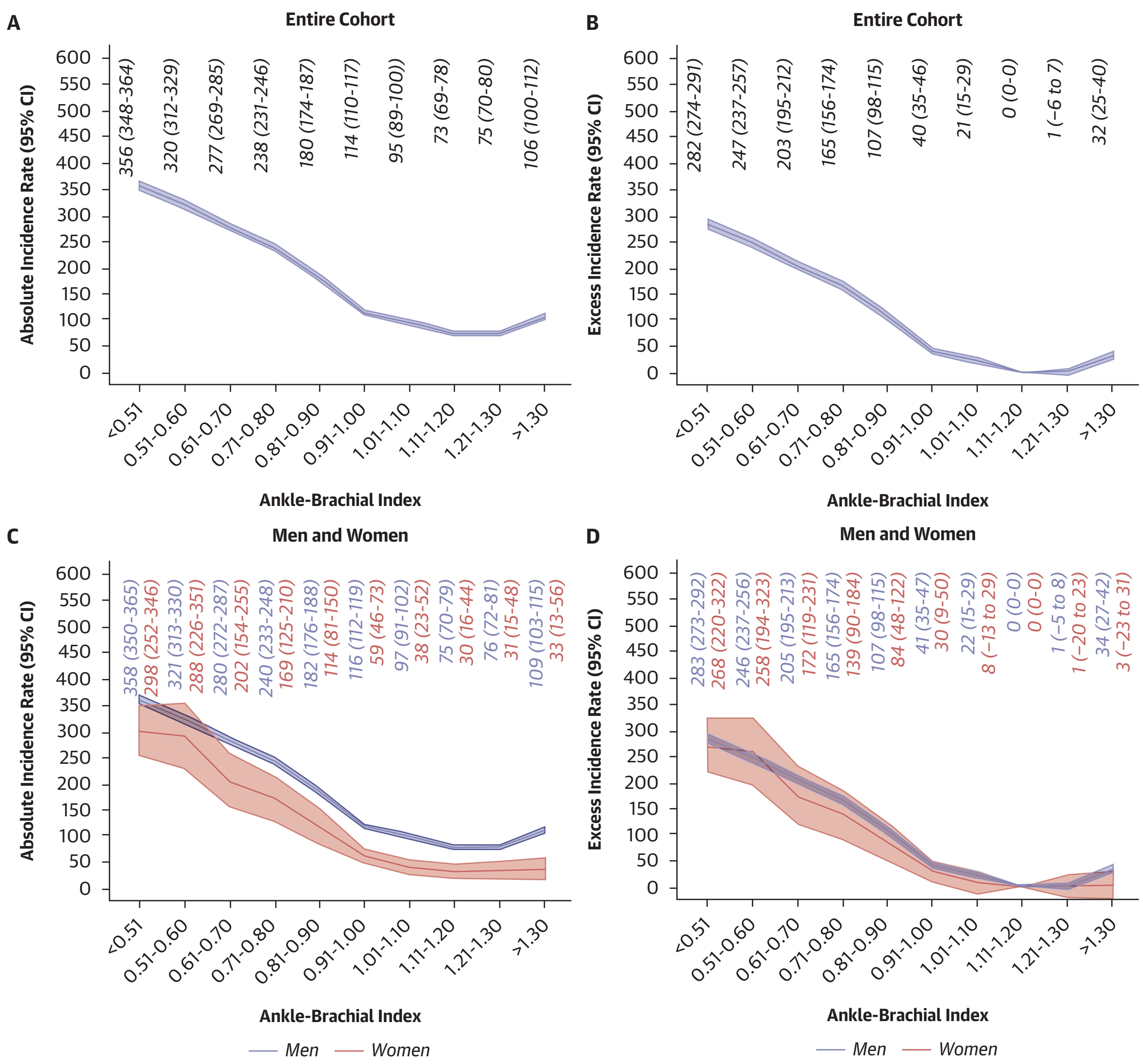
FIGURE 1 Risk of Major Adverse Limb Events by Ankle-Brachial Index



HRs and 95% CIs derived from Cox proportional hazards regression models for risk of major adverse limb events with ankle-brachial index modeled as a continuous variable using restricted cubic splines. (A) Total study population. Models adjusted for age, sex, race/ethnicity, Veterans Integrated Service Network, smoking history, prevalent cardiovascular disease, diabetes, antihypertensive therapy, systolic blood pressure, diastolic blood pressure, statin therapy, antiplatelet therapy, body mass index, estimated glomerular filtration rate, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, and triglycerides. (B and C) Stratified by sex. Models adjusted for all the above covariates except for sex. (D and E) Stratified by race. Models adjusted for all the above covariates except for race/ethnicity.

FIGURE 2 10-Year Absolute Incidence Rate and Excess Incidence Rate of Major Adverse Limb Events by Ankle-Brachial Index



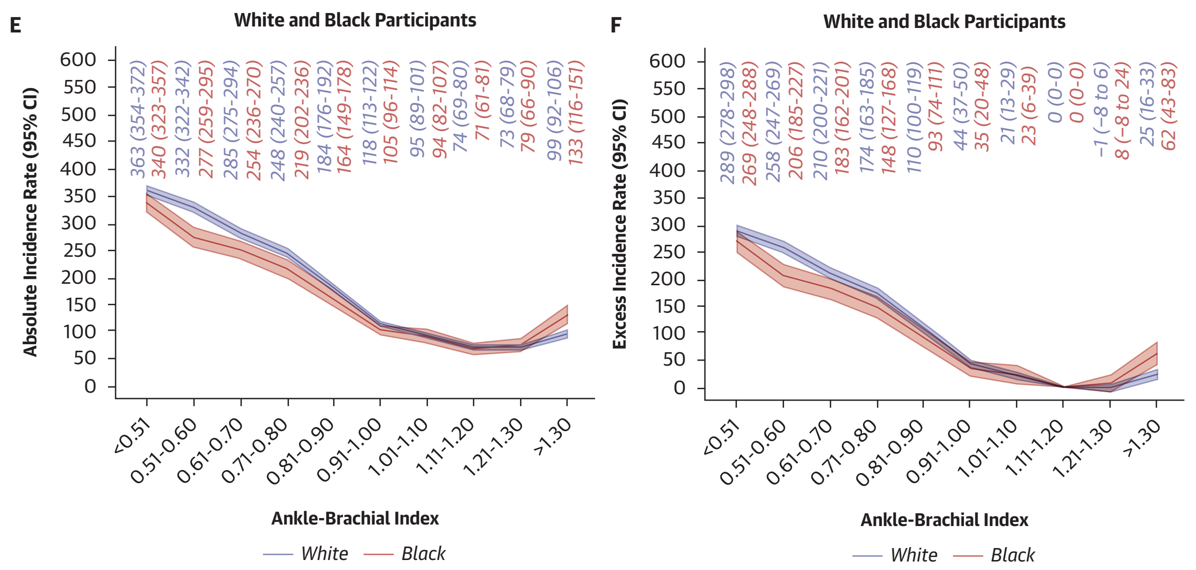
(A) Absolute 10-year incidence rate per 1,000 persons of major adverse limb events in the full cohort. (B) Excess 10-year incidence rate per 1,000 persons of major adverse limb events in the full cohort. (C and D) Absolute and excess risk per 1,000 persons stratified by sex. (E and F) Absolute and excess risk per 1,000 persons stratified by White and Black race.

Continued on the next page

Table 3). Even those with a borderline ABI of 0.91 to 1.00 had an increased incidence compared with those in 1.11-1.20 category (incidence rates [IR] per 1,000 person-years: 15.0 [95% CI: 14.5-15.5] and 9.3 [95% CI: 8.8-9.9], respectively). Similar trends were also seen for both men and women (Supplemental Table 4). Incidence of MALE was lower in women than men in each ABI category. The highest age-adjusted incidence rates were for men and women with an ABI <0.50 (IR per 1,000 person-years: 72.8 [95% CI: 71.0-74.7] and 56.9 [95% CI: 47.8-67.8], respectively). Similar patterns were seen for major amputation, total amputation, and revascularization in the total population (Supplemental Table 3). Rates of major amputation, total amputation, and revascularization were numerically higher for men than women in all ABI categories (Supplemental Table 4).

When stratified by race, age-adjusted incidence rates of MALE similarly increased as ABI category decreased from the reference range in both White

FIGURE 2 Continued



and Black individuals (Supplemental Table 5). Black individuals had numerically higher age-adjusted incidence rates of major amputation and total amputation compared with White individuals within every ABI category. This pattern was reversed for age-adjusted incidence rates of revascularization.

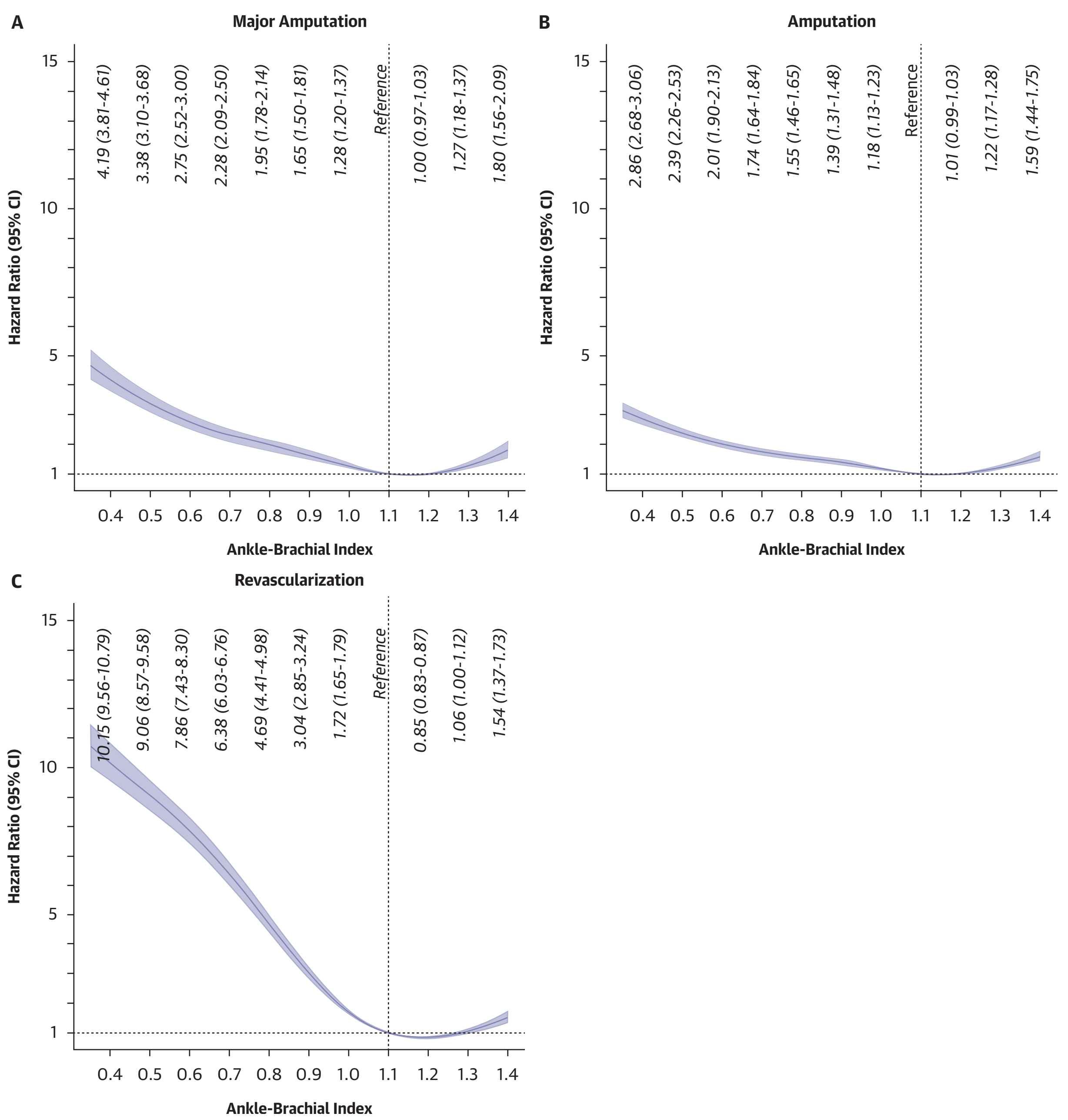
The risk of MALE in the total population is reported with ABI modeled as a continuous variable (**Figure 1A**) and a categorical variable (Supplemental Figure 2A). Risk followed an inverse j-shaped distribution with the greatest risk of MALE associated with the lowest ABI values. This risk was elevated even for individuals with an ABI of 1.0 compared with those in the reference group with an ABI of 1.1 (multivariable-adjusted HR [HR_{adj}]: 1.38 [95% CI: 1.34-1.43]). There was a similar pattern for those within a low-normal ABI category of 1.01 to 1.10 or borderline ABI category of 0.91 to 1.00 compared with those with a reference ABI of 1.11 to 1.20 (HR_{adj}: 1.25 [95% CI: 1.16-1.36]; and HR_{adj}: 1.53 [95% CI: 1.43-1.64], respectively). Results were similar in sensitivity analyses when follow-up began at the time of qualifying ABI (Supplemental Figure 3). When the strengths of association between each covariate in the regression model and incident MALE were ranked by chi-square statistic minus df, ABI was ranked highest (Supplemental Figure 4A).

Similar risk distributions were seen when participants were stratified by sex (**Figures 1B and 1C**, Supplemental Figures 2B and 2C) or race (**Figures 1D and 1E**, Supplemental Figures 2D and 2E). An ABI of 1.0 compared with 1.1 was again associated with an increased risk of MALE for men (HR_{adj}: 1.38 [95% CI: 1.34-1.42]), women (HR_{adj}: 1.58 [95% CI: 1.25-2.01]), White individuals (HR_{adj}: 1.41 [95% CI: 1.36-1.46]), and Black individuals (HR_{adj}: 1.31 [95% CI: 1.21-1.41]). Similar associations were present for a borderline ABI category of 0.91 to 1.00 compared with a reference ABI of 1.11 to 1.20.

The 10-year cumulative incidence rates for MALE by ABI followed a similar inverse j-shaped distribution (**Figure 2A**). Compared with an ABI 1.11 to 1.20, those with a borderline ABI of 0.91 to 1.00 had a 4% absolute increase in MALE risk at 10 years. This corresponded to an excess of 40 MALE events (95% CI: 35-46) per 1,000 persons over 10 years (**Figure 2B**). Similar incidence rate curves were seen when participants were stratified by sex (**Figures 2C and 2D**) or race (**Figures 2E and 2F**) with men exhibiting a numerically higher absolute risk in each ABI category. For the 10-year absolute incident rate of all-cause mortality before MALE, borderline ABI (range 0.91-1.00) was associated with an excess of 51 (95% CI: 41-62) deaths per 1,000 persons compared with normal ABI (range 1.11-1.20). Stratifying by sex or race showed similar trends (Supplemental Figure 5).

Risk of lower extremity major amputation and total amputation followed an inverse j-shaped distribution for the total population (**Figures 3A and 3B**, Supplemental Figures 6A and 6B). When ABI was treated as a categorical variable, men with a

FIGURE 3 **Risk of Different Limb Outcomes by Ankle-Brachial Index**



HRs and 95% CIs derived from Cox proportional hazards regression models for risk of (A) lower extremity major amputation, (B) total amputation, or (C) revascularization within the total study population with ABI modeled as a continuous variable. Models adjusted for age, sex, race/ethnicity, Veterans Integrated Service Network, smoking history, prevalent cardiovascular disease, diabetes, antihypertensive therapy, systolic blood pressure, diastolic blood pressure, statin therapy, antiplatelet therapy, body mass index, estimated glomerular filtration rate, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, and triglycerides.

borderline ABI of 0.91 to 1.00 had an increased risk of major amputation (HR_{adj}: 1.35 [95% CI: 1.18-1.55]) and total amputation (HR_{adj}: 1.22 [95% CI: 1.12-1.34]) compared with an ABI of 1.11 to 1.20 (Supplemental Table 6). Major amputations and total amputations in women were rare, and the associations between ABI category and risk were statistically significant only for very low ABI values. When

TABLE 2 Risk of Lower Extremity Outcomes by Race and ABI Category

ABI	Major Amputations/n	Multivariable-Adjusted HR (95% CI)[a]	Total Amputations/n	Multivariable-Adjusted HR (95% CI)[a]	Revascularizations/n	Multivariable-Adjusted HR (95% CI)[a]
White (n = 165,975)						
≤0.50	769/14,963	4.15 (3.53-4.88)	1,283/14,963	2.75 (2.48-3.05)	3,920/14,963	11.24 (10.11-12.50)
0.51-0.60	433/11,971	2.91 (2.44-3.46)	828/11,971	2.18 (1.95-2.44)	2,861/11,971	9.77 (8.77-10.88)
0.61-0.70	439/14,019	2.48 (2.08-2.94)	816/14,019	1.77 (1.59-1.98)	2,763/14,019	7.70 (6.92-8.58)
0.71-0.80	357/16,130	1.97 (1.65-2.35)	755/16,130	1.56 (1.40-1.74)	2,257/16,130	6.28 (5.64-7.01)
0.81-0.90	339/14,417	1.83 (1.53-2.19)	718/14,417	1.45 (1.30-1.62)	1,530/14,417	4.20 (3.76-4.70)
0.91-1.00	378/31,157	1.42 (1.21-1.68)	1,370/31,157	1.25 (1.13-1.38)	1,755/31,157	2.22 (1.99-2.48)
1.01-1.10	198/14,299	1.10 (0.90-1.35)	524/14,299	1.06 (0.94-1.19)	557/14,299	1.58 (1.39-1.80)
1.11-1.20	192/17,055	Ref	545/17,055	Ref	384/17,055	Ref
1.21-1.30	209/17,668	1.11 (0.91-1.35)	582/17,668	1.06 (0.95-1.20)	313/17,668	0.86 (0.74-0.99)
>1.30	242/14,296	1.51 (1.24-1.82)	639/14,296	1.37 (1.22-1.54)	392/14,296	1.45 (1.26-1.67)
Black (n = 42,173)						
≤0.50	324/4,582	3.77 (2.87-4.95)	471/4,582	2.80 (2.31-3.41)	1,013/4,582	9.08 (7.38-11.33)
0.51-0.60	187/3,612	2.75 (2.06-3.66)	286/3,612	2.13 (1.73-2.63)	631/3,612	6.96 (5.54-8.72)
0.61-0.70	191/4,224	2.36 (1.78-3.15)	291/4,224	1.78 (1.45-2.19)	670/4,224	6.11 (4.88-7.65)
0.71-0.80	136/4,850	1.75 (1.30-2.36)	259/4,850	1.68 (1.36-2.07)	481/4,850	4.77 (3.80-6.00)
0.81-0.90	138/3,790	1.93 (1.44-2.60)	232/3,790	1.60 (1.29-1.98)	294/3,790	3.18 (2.50-4.04)
0.91-1.00	176/7,616	1.17 (0.88-1.56)	340/7,616	1.13 (0.93-1.38)	354/7,616	1.84 (1.46-2.33)
1.01-1.10	64/3,364	1.03 (0.73-1.45)	139/3,364	1.10 (0.87-1.39)	122/3,364	1.51 (1.15-1.99)
1.11-1.20	66/3,920	Ref	138/3,920	Ref	88/3,920	Ref
1.21-1.30	65/3,539	1.08 (0.77-1.53)	152/3,539	1.22 (0.97-1.54)	62/3,539	0.80 (0.58-1.11)
>1.30	96/2,676	2.00 (1.46-2.74)	178/2,676	1.80 (1.44-2.25)	117/2,676	2.08 (1.58-2.75)

[a]Fully adjusted model includes the following covariates: age, sex, Veterans Integrated Service Network, smoking history, prevalent cardiovascular disease, diabetes, antihypertensive therapy, systolic blood pressure, diastolic blood pressure, statin therapy, antiplatelet therapy, body mass index, estimated glomerular filtration rate, low-density lipoprotein cholesterol, and high-density lipoprotein cholesterol.

ABI = ankle-brachial index.

stratified by race, a borderline ABI of 0.91 to 1.00 was also associated with an increased risk of major amputation (HR_{adj}: 1.42 [95% CI: 1.21-1.68]) but not total amputation (HR_{adj}: 1.25 [95% CI: 1.13-1.38]) compared with an ABI of 1.11 to 1.20 for White individuals (**Table 2**). This risk association did not reach statistical significance for Black individuals (HR_{adj}: 1.17 [95% CI: 0.88-1.56] and HR_{adj}: 1.13 [95% CI: 0.93-1.38] vs ABI 1.11-1.20 for major and total amputations, respectively). Diabetes was the covariate ranked highest in relative importance for both major amputation (Supplemental Figure 4B) and total amputation (Supplemental Figure 4C) in the total population.

Risk of lower extremity revascularization also followed an inverse j-shaped distribution (**Figure 3C**, Supplemental Figure 6C). When ABI was treated as a categorical variable, men with borderline ABIs of 0.91 to 1.00 or low-normal ABIs of 1.01 to 1.10 were at increased risk of undergoing revascularization compared with those with ABIs of 1.11 to 1.20 (HR_{adj}: 2.03 [95% CI: 1.85-2.24] and HR_{adj}: 1.53 95% CI: 1.36-1.71], respectively) (Supplemental Table 7). These associations were more pronounced for women (HR_{adj}: 5.08 [95% CI: 1.83-14.11] and HR_{adj}: 3.61 [95% CI: 1.21-10.82], respectively) (Supplemental Table 7). A similar pattern was seen for both White and Black individuals (**Table 2**). ABI ranked highest among all covariates for its relative importance for revascularization in the total population (Supplemental Figure 4D).

DISCUSSION

In a large, racially diverse cohort of men and women without baseline PAD, there was a range of limb risk across the full spectrum of ABI values (**Central Illustration**). Even borderline ABI values and low-normal ABI values were markers of increased MALE risk. When examined separately, this risk persisted among men, women, White individuals, and Black individuals. Similar trends were seen for lower extremity major amputation and revascularization among subgroups. A borderline ABI of 0.90 to 0.99 was not benign; compared with an ABI of 1.10 to 1.19, it was more strongly associated with incident MALE than reduced estimated glomerular filtration rate, current smoking, and established CVD.

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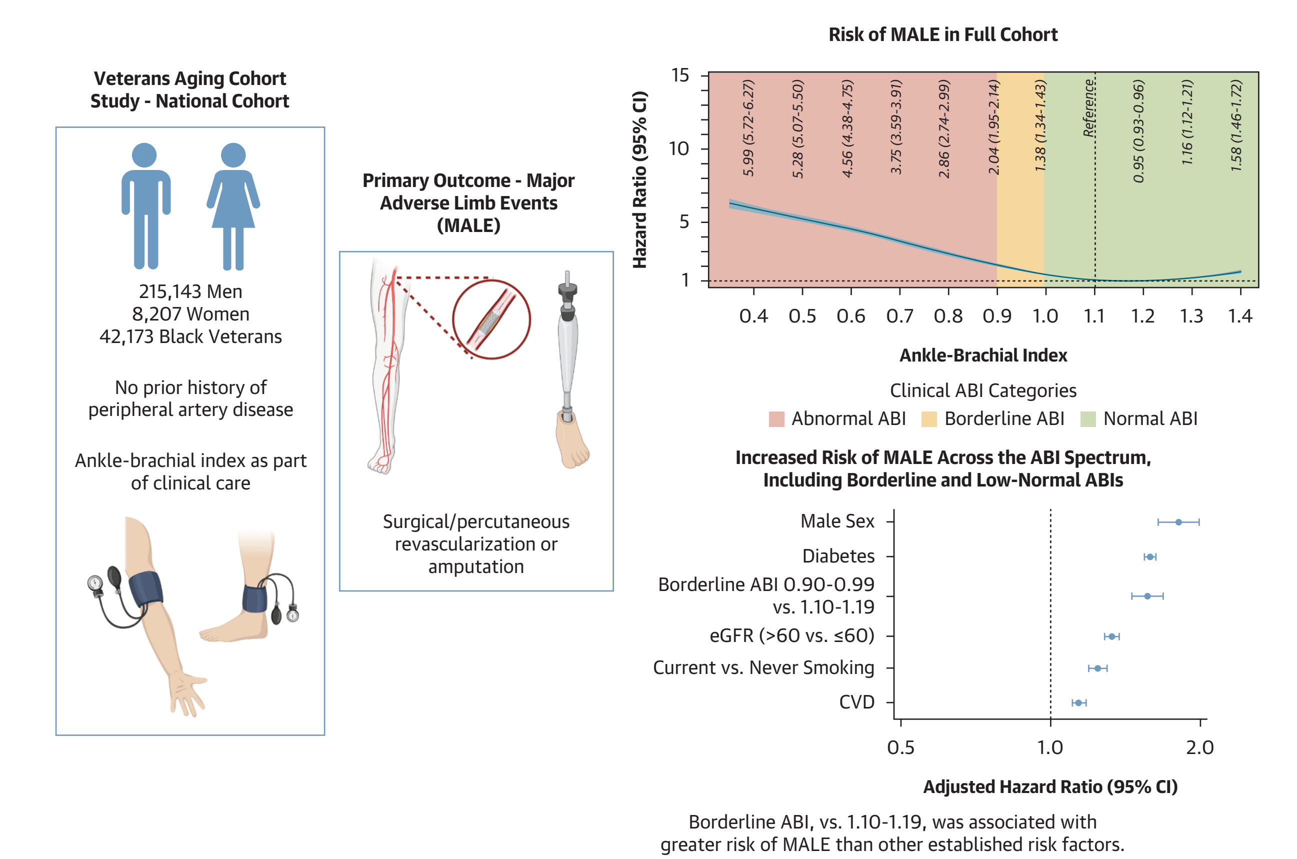
CENTRAL ILLUSTRATION Overview of Cohort and Primary Study Outcomes



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We present the HRs for selected covariates to put the ankle-brachial index (ABI) in context. The study was designed to estimate the effects of the ABI rather than the covariates; thus, this is for illustrative purposes only. Created in BioRender. eGFR = estimated glomerular filtration rate; MALE = major adverse limb events.

For decades, clinicians have reassured patients with borderline ABI values that they were nearly normal. These data suggest otherwise. The ABI, rather than treated as a binary test, is better understood as a continuum—one that captures the earliest signals of limb threat. Recognizing this continuum offers an opportunity for earlier prevention and a new respect for a measure hiding in plain sight.

While several prior studies have demonstrated an association between low ABI and poor limb outcomes, most have been restricted to individuals with known PAD,[10,11,34] have focused on extreme ABI measures,[10] or have not distinguished between major amputation and diabetes-associated minor amputation.[35] In the ARIC (Atherosclerosis Risk in Communities) study, investigators found an increased risk of total amputation in those with a low-normal ABI of 1.01 to 1.10 (HR_{adj}: 1.73 [95% CI: 1.07-2.79]) compared with an ABI 1.11 to 1.20,[9] where we found no significant association with major amputation or total amputation in the total population in the present study (HR_{adj}: 1.11 [95% CI: 0.94-1.31] and HR_{adj}: 1.08 [95% CI: 0.97-1.20], respectively) despite markedly lower amputation rates in ARIC. There are several possible explanations for the stronger risk associations in ARIC. First, the ABI measures in ARIC were collected from 1987-1989 compared with 2000-2021 in the VACS-National Cohort, and differences in lifestyle and primary prevention efforts may account for some of these differences. ABI measures in ARIC were only performed in 1 leg, and inclusion of both legs in our analysis may have led to more accurate ABI categorization. The ABI was measured using an oscillometric device in ARIC, while our study used ABI measures from clinical practice. This is more commonly done with an integrated vascular laboratory Doppler system that may have resulted in more accurate ABI measures. Further, we did demonstrate an association between borderline ABI of 0.91 to 1.00 vs 1.11 to 1.20 and major amputation risk in the total population, further highlighting the misleading nature of a "borderline" ABI classification.

The limb risk associated with higher ABI in the present study is also notable. Compared with an ABI 1.11 to 1.20, an ABI >1.30 was associated with a 50% increased risk of MALE. The mechanisms underlying this risk are multifactorial. Although advanced age, diabetes, smoking, and chronic kidney disease are well established risk factors for arterial stiffness, the risk association in the present study persisted after adjusting for these covariates.[36] These modest increases in ABI may be caused by median arterial calcification, an important component of PAD pathogenesis that can contribute to arterial narrowing and thrombus formation independent of atherosclerotic plaque.[37,38] Current guidelines do not assign an "abnormal" label for an elevated ABI until it exceeds 1.40,[2] further emphasizing the inadequacy of our clinical terminology for capturing the continuum of ABI risk.

Several additional unique aspects of the present study are worth noting. First, given our large sample size, we were able to examine rates and risk across race subgroups. Regardless of ABI, Black individuals had higher rates of amputation compared with White individuals, and further studies are needed to better understand these racial differences.[39] Second, to our knowledge, this is the first analysis of ABI and incident revascularization as a separate endpoint in individuals free of prior clinical PAD. Many individuals in the normal ABI range later underwent revascularization, and whether this was caused by low sensitivity of the ABI,[40] subsequent limb thromboembolism,[41] or prioritization of arterial anatomy in guiding revascularization[42] remains unclear. Finally, given our sample size, we were adequately powered to examine the risk of MALE across the full spectrum of ABI values. In contrast to analyses from ARIC, which found no increased risk of amputation with an ABI >1.30,[9] we found a 1.8-fold increased risk of major amputation, 1.6-fold increased risk of total amputation, and 1.5-fold increased risk of revascularization in those with ABI of 1.4 vs 1.1 even after accounting for diabetes and renal function, further highlighting the continuum of ABI risk.[10,43]

STUDY LIMITATIONS. First, our study population was predominantly male, and we had much lesser power for the analyses among women. Although the number of women in this study (n = 8,207) exceeds those found in some other reports on this topic,[10] even larger studies are needed to validate these results. Second, our ABI studies were performed as part of routine clinic care rather than as a population-based screening measure, and it is unknown whether the same risk associations would be seen for screening ABI values. Third, our study did not include details regarding anatomic distribution of PAD or technical details of revascularization procedures. Fourth, we do not have details on whether oscillometric vs Doppler ABI measures performed differently in terms of risk prediction. Finally, because this was an observational study, there is a possibility of residual confounding beyond what was accounted for in our statistical models, including any potential impact of aspirin therapy on MALE.

CONCLUSIONS

In this prospective study of more than 220,000 veterans and over 28,000 adverse limb events, values across the ABI continuum, rather than a cutoff, were associated with limb risk. These trends did not vary by sex or race. ABI was one of the strongest markers of future MALE risk. A "borderline" ABI is not benign, and our findings suggest this terminology falsely reassures patients and clinicians. Future studies should explore potential mechanisms linking modest reductions in ABI and adverse outcomes as well as test the impact of alternative ABI thresholds on clinical outcomes.

DATA AVAILABILITY Access to VA/CMS data was provided by the Department of Veterans Affairs, Veterans' Health Administration, Office of Research and Development, Health Services Research and Development, and the VA Information Resource Center under Project Numbers SDR 02-237 and 98-004. This work uses data provided by patients and collected by the VA as part of their care and support.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This work was supported by funding from the National Institutes of Health K23 HL151871, P01 AA029545, and U24 AA020794. The views and opinions expressed in this manuscript are those of the authors and do not necessarily represent those of the Department of Veterans Affairs or the U.S. government. Dr Aday has received consulting fees from Merck; and has received speaker fees from Tactile Medical, all outside of the current work. Dr Beckman has received consulting fees from JanOne, Janssen, and Novartis; and has received research funds from Bristol Myers Squibb, all outside of the current work. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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KEY WORDS amputation, ankle-brachial index, peripheral artery disease, revascularization

APPENDIX For supplemental tables and figures, please see the online version of this paper.